

WED-Jan-29-2025: Rocket optimization & Examples

We have $\sum(\dots) = 0$, the most rational conclusion is that individual terms in $\sum \dots$ are zero, so $(\dots) = 0$.

$$\Rightarrow \frac{1}{\lambda_i(1+\lambda_i)} + \alpha g_e I_{sp_i} \frac{(\epsilon_i - 1)}{(1+\lambda_i)(\epsilon_i + \lambda_i)} = 0 \quad (C)$$

We can solve (C) for $\lambda_i \Rightarrow \lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1 - \epsilon_i) - 1} \quad (D)$

(D) gives the λ_i that maximizes $\frac{M_L}{M_0}$ considering $\Delta u_{b,imp} = \text{cte}$, but we need an extra equation for α !

α is determined from known $\Delta u_{b,imp}$:

remember: $\Delta u_{b,imp} = \sum_{i=1}^n g_e I_{sp_i} \ln\left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i}\right)$

known

Put λ_i from (D) in the above Eqⁿ & solve it for α , after a lot of math calculations, we get:

$$\left\{ \frac{1}{\epsilon_1} \left[1 - \frac{1}{\alpha g_e I_{sp_1}} \right] \right\}^{\frac{g_e I_{sp_1}}{\Delta u_{b,imp}}} \times \left\{ \frac{1}{\epsilon_2} \left[1 - \frac{1}{\alpha g_e I_{sp_2}} \right] \right\}^{\frac{g_e I_{sp_2}}{\Delta u_{b,imp}}} \times \dots \times \left\{ \frac{1}{\epsilon_n} \left[1 - \frac{1}{\alpha g_e I_{sp_n}} \right] \right\}^{\frac{g_e I_{sp_n}}{\Delta u_{b,imp}}} = e \rightarrow 2.71828 \quad (E)$$

term.

(1) You should write for n (number of stages) the { }

(2)

(1-5)

We use MATLAB to solve (E), get α , then use (D) to get λ_i

Example: Reconsider the previous 3-stages rocket example:

$$I_{sp_1} = I_{sp_2} = I_{sp_3} = 370 \text{ s}, \quad \epsilon_1 = 0.05, \quad \epsilon_2 = 0.0698, \quad \epsilon_3 = 0.1892$$

$$\frac{M_L}{M_o} = \frac{68,000}{2,500,000} = 0.0272, \quad \lambda_1 = 0.3021, \quad \lambda_2 = 0.4464$$

$$\lambda_3 = 0.6126, \quad \Delta u_{b,imp} = 11023 \frac{\text{m}}{\text{s}} \quad (\text{these data are given in last example}).$$

Let us optimize the rocket to maximize $\frac{M_L}{M_o}$.

Considering $\epsilon = \text{cte}$, $\Delta u_{b,imp} = \text{cte}$, $I_{sp} = \text{cte}$, λ changes $\rightarrow$

$\frac{M_L}{M_o}$ will change as well.

Solution: Solve (E) with MATLAB code to get α_{ge} , then find λ_i from (D), then find $\left(\frac{M_L}{M_o}\right)_{opt} = \frac{\lambda_1}{1+\lambda_1} \times \frac{\lambda_2}{1+\lambda_2} \times \frac{\lambda_3}{1+\lambda_3}$

MATLAB code gives solution for (E) as $\alpha_{ge} = 0.003554$

the code gave also λ_i , as $\lambda_1 = 0.2005$, $\lambda_2 = 0.3125$, $\lambda_3 = 0.4692$

$$\left(\frac{M_L}{M_o}\right)_{opt} = 0.0294 > 0.0272$$

check that total $\Delta u_{b,imp}$ remains constant.

$$\Delta u_{b,imp,i} = u_{eq} \ln \frac{1+\lambda_i}{\epsilon_i + \lambda_i} \quad (\text{for } HW \neq 2, Q-1, \text{ use } u_{eq,i} = \frac{\Delta u_{b,i}}{\ln \frac{M_{o,i}}{M_{o,i}-M_i}})$$

(2)

(1-5)

We compute $\Delta u_{b,imp,i} \rightarrow \Delta u_{b,imp,1} = 5688.1 \frac{m}{s}$

$$\Delta u_{b,imp,2} = 4477.2 \frac{m}{s}$$

$$\Delta u_{b,imp,3} = 857.7 \frac{m}{s}$$

$$\Delta u_{b,imp,tot} = \sum_{i=1}^3 \Delta u_{b,imp,i} = 11023 \frac{m}{s} \rightarrow \text{Same as before}$$

We have optimized $\frac{M_L}{M_{o1}}$, assume M_L is the same as before,

$M_L = 68,000 \text{ kg} \rightarrow M_{o1}$ reduces in optimized case:

$$M_{o1,opt} = \frac{M_L}{\left(\frac{M_L}{M_{o1}}\right)_{opt}} = \frac{68 \text{ k}}{0.0294} = 2.313 \times 10^6 \text{ kg}$$

dropped from $2.5 \times 10^6 \text{ kg}$

on the other hand, if $M_o = 2.5 \times 10^6 \text{ kg} = \text{etc}$ as before, M_L increases:

$$M_{L,opt} = M_{o1} \left(\frac{M_L}{M_{o1}}\right)_{opt} = 2.5 \times 10^6 \times (0.0294) = 73,500 \text{ kg}$$

The payload mass improved from 68 k to 73.5 k kg!